Lucas C. Cordeiro
Department of Computer Science, University of Manchester, UK
https://research.manchester.ac.uk/en/persons/lucas.cordeiro

Employment History:

Professor (since 09/2024), University of Manchester, UK

Associate Professor (5/2018-08/2024), University of Manchester, UK

Director of Business Engagement and Innovation (since 09/2024), University of Manchester, UK

Director of the Arm Center of Excellence (since 6/2021), University of Manchester, UK

Post-doctoral Researcher (7/2016-04/2018), University of Oxford, UK

Assistant Professor (9/2009-06/2016), Federal University of Amazonas, Brazil

Software Engineer (01/2005-8/2008), Siemens Mobile and CT-PIM/NXP, Manaus, Brazil

Education:

PhD in Computer Science, 5/2011, University of Southampton, UK
Thesis: *SMT-Based Bounded Model Checking of Multi-threaded Software in Embedded Systems*
(Supervisor: Prof. Dr. B. Fischer).

MSc in Informatics, 8/2007, Federal University of Amazonas, Brazil
Thesis: *An Agile HW/SW Development Methodology for Embedded Systems*
(Supervisor: Prof. Dr. R. Barreto)

Research Interests:

Algorithms, software engineering, formal methods, automated program verification, synthesis, and testing, concurrency, security, LLM-based verification, and embedded & cyber-physical systems.

Professional Activities:

200 reviewed publications, incl. 44 journal and 156 conference papers, h-index 37, **Amazon Research Award** (2025), **Paper awards** at ICWS'26, ASE'24, ASE'23, SBSeg'23, SBESC'15, ICSE'11, SAC'08.

56 awards from the intl competition on software testing (Test-Comp) and verification (SV-COMP) 2012-2026.

14 PhD, 3 MPhil and **35 MSc students graduated**, 5 current PhD students; **4 CVEs** found (AWS, GifLib).

Proven track record of securing **research funding** from the government (EPSRC, UKRI, Innovate UK, European Commission, GCHQ, British Council, Royal Society, CAPES, CNPq, FAPEAM) and industry (Amazon, BAE Systems, Intel, Motorola, Samsung, Nokia, Arm) (career total in excess of USD16.4M).

URL of **Google Scholar** (https://scholar.google.com/citations?user=Lje1SFgAAAAJ&hl=en&oi=ao) and **DBLP** (http://dblp.uni-trier.de/pers/hd/c/Cordeiro:Lucas_C=).

Selected Relevant Publications to this Proposal:

Y. Sun, J. Liu, H. Alqasimi, **L. Cordeiro**, D. Kroening, J. Xue. "Agentic Model Checking". *ICSE*, 2027.

C. Wei, S. Cai, Y. Charalambous, T. Wu, S. Godboley, **L. Cordeiro**. "VeriExploit: Automatic Bug Reproduction in Smart Contracts via LLMs and Formal Methods". *ASE*, pp. 1415–1426, 2025.

M. Ferrag, A. Battah, N. Tihanyi, R. Jain, et al., **L. Cordeiro**. "SecureFalcon: Are We There Yet in Automated Software Vulnerability Detection With LLMs?". *IEEE Trans. Softw. Eng.*, 51(4), pp. 1248–1265, 2025.

N. Tihanyi, T. Bisztray, M. Ferrag, R. Jain, **L. Cordeiro**. "How secure is AI-generated code: a large-scale comparison of large language models". *Empir. Softw. Eng.*, 30(2):47, 2025.

M. Pirzada, G. Reger, A. Bhayat, **L. Cordeiro**. "LLM-Generated Invariants for Bounded Model Checking Without Loop Unrolling". *ASE*, pp. 1395–1407, 2024. **ACM Distinguished Paper Award**.

L. Cordeiro, B. Fischer. "Verifying Multi-threaded Software using SMT-based Context-Bounded Model Checking". *ICSE*, pp. 331–340, 2011. **ACM Distinguished Paper Award**.

L. Cordeiro, B. Fischer, J. Marques-Silva. "SMT-Based Bounded Model Checking for Embedded ANSI-C Software". *ASE*, pp. 137–148, 2009. **Most Influential Paper Award**.